

COMPUTATIONAL RECORD FOR REVISION R34

QIAN QI
Peking University

1. SUBMISSION IDENTITY AND PROVENANCE

R34 is based on frozen parent 88015b77a0c26e7b883156410b684b229fd7e86f. Its isolated branch is revision/econometrica-r34-verified-policy-frontier-2026-09-24. The source R30 report at e9bc144fbb6843d6a3436825a28584efb64fb8f1 reviews manuscript 3394815f4ccbf5917582c8f756b535adf87783cc. R31 and R32 are intervening revisions, not new referee reports. The frozen parent contains an executed R32 publication and a staged R33 source archive. It does not contain an executed R33 publication. We distinguish those objects explicitly.

The repository snapshot was obtained from a successful source-packaging run, 35979134670, at new-branch commit 2a5f133a4f352521a00c8763356c1067b5bb4f90. The R33 source archive was decoded locally and inspected; its mathematical support construction was then executed anew as part of R34. Both the original transfer bytes and the R32 executed evidence remain unchanged. The restart-propagation protocol was committed at 70526abec87eeadd52dd7b30eaf75a78ba70e79c before implementation or evaluation of that addition. At that point the new execution of the support portfolio was already available. The protocol does not relabel known results as prospective observations.

2. NUMERICAL OBJECTS AND PROVENANCE BOUNDARIES

The unchanged model has horizons 4,8,12, tolerances 1/100,1/20, and seven installed rules at each horizon. The exact compiled incumbent files are the frozen R32 inputs. No neural or spline fitting is rerun. The support and local multiplier sets are both $\{0, 1, 4, 16, 64, 256, 1024, 4096\}$, the witness accuracy parameter is 1/10000, and the representation cap is 200,000 pieces. Installed-rule fitting records remain inherited measurements; they are not added to new-machine timings to manufacture a cold-start speedup.

The support stage constructs 168 unrestricted support objects, 42 freshly reconstructed class controls, and 42 primal–dual summaries. Every class control is rebuilt from primitives on the new machine, not treated as a free inherited solution. Its complete rational object must equal the inherited comparator, which supplies a reproducibility check but not its construction. The staged R33 sources contain a description of an earlier timing omission. R34 preserves those source bytes as history; it does not claim to have rerun or reconstructed an unavailable earlier R33 execution. The implemented complete-cost accounting is used from the first R34 support execution.

The new backward propagation constructs another 42 certificates. It does not train a model, alter the feasible deployment, or solve another unrestricted support problem. It builds local action minima and outer bound maxima from the verified primitives and existing witnesses. All local multipliers, all unsuccessful feasibility candidates in the support portfolio, all unchanged propagation outcomes, and all positive remaining gaps are retained.

3. INDEPENDENT VERIFICATION AND TESTS

The final read-only audit reloads 294 serialized objects: 168 support certificates, 42 classical controls, 42 primal–dual summaries, and 42 restart-propagated objects. It disables optimizing envelope routines before checking the complete portfolio and propagated identities. Hashes bind the exact files; all Bellman inequalities, selected equalities, own-policy values, feasible operating envelopes, lower-bound functions, and reported integrals are recomputed with rational arithmetic. Every affine interval is checked through both endpoint limits, and actual isolated knot values are separately checked. The arithmetic library is shared with the constructor; this is a read-only verifier separated from optimization, not a formally verified arithmetic kernel.

Eight malformed support tests, eight malformed primal–dual tests, and twelve malformed propagation tests are required to fail. The support test compares each of four rational restart states with all 27 contingent two-period plans. The additional propagation test enumerates 108 such state–plan combinations and 1,512 equal mixtures, and tests all satisfying the conditional all-restart restriction. Eligible counts are reported, including 4 deterministic plans and 20 mixtures. These finite tests supplement rather than replace the continuum proof and serialized whole-cell comparisons.

The inherited active-boundary rational inequalities are also rerun. They verify the interval-wide positive derivative constants and the upper-exit error bound used in the scalar optimum theorem. The endpoint payoff enclosure remains inherited from the original validated quadrature; it is not described as newly recomputed. No result of the maintenance study changes the original all-domain stopped-economy bound or supplies the missing current-state residual bridge.

4. COMPLETE COMPUTATIONAL ACCOUNTING

Support-stage timings include upper and lower witness construction, all support solves, exact own-policy evaluations, freshly rebuilt class controls, comparison and selection, encoding, and internal checks. Per-configuration totals charge shared portfolios at both tolerances; the summary separately reports unique batch work. The diagnostic exact operating dynamic program is recorded separately and is not an input to the certificate constructor. The first support audit and the final combined audit are separate executed checks, and their command-level wall-clock ledgers are retained.

The new propagation’s construction, internal checks, and encoding total 178.896 seconds. Its read-only additional audit takes 90.990 seconds; the full final base-plus-propagation audit takes 154.340 seconds. These numbers are descriptive measurements on the recorded machine. They do not estimate an average over independent machine runs. All intermediate and output costs remain additional to, not substitutes for, the inherited fitting costs of the already installed rule.

The largest stored propagated lower bound has 471 pieces; closure and local-minimum rationals reach 716 bits. Compressed closure files occupy 17.846 MiB. Raw bytes, compressed bytes, checksums, piece counts, and bit lengths are individually recorded. Stored proof size is not peak process or intermediate-overlay memory. The one-dimensional computation does not establish a dimension-free scalability rate.

5. REPRODUCTION AND PUBLICATION

From the repository root, with Python without the `-O` optimization flag, run:

```
R=revisions/2026-09-24-r34
python "$R/run_study.py"
python "$R/run_closure.py"
python "$R/replication/report_global.py" "$R"
```

```
python "$R/replication/report_restart.py" "$R"
python revisions/2026-09-24-r32/replication/boundary.py \
  --output "$R/results/boundary_recheck.json"
python "$R/assemble.py"
bash "$R/build.sh"
python "$R/publication.py"
```

Only the Python standard library is required for the newly executed exact computations. The inherited fitting procedures have their own recorded dependencies and are not called. The publication workflow reconstructs sources from a checksummed archive, checks the frozen inputs, executes the full computation, builds the article, supplement, response, computation record, and historical annex, and commits the resulting files only to the isolated R34 branch. The source commit, workflow run, publication hashes, and exact certificate hashes are recorded separately. No file attempts to embed the hash of its own future Git commit.

6. PRESERVATION AND INTERPRETATION

The historical annex contains the complete R32 article, supplement, response, computation record, and its existing historical annex, with unchanged pages and file hashes. The current article and supplement also retain the previous source text after input-path changes, with new theory and evidence inserted. The staged R33 transfer archive remains untouched. The preservation map records parent identity, source hashes, all insertions, and inherited document hashes.

All 42 feasible deployments preserve installed operating values. Eighteen installed-neural configurations have strictly lower intervention cost than the conservative class control. The new propagation tightens 22 global intervals without changing any deployment; twelve spline cases retain exact zero global gap. No positive gap closes in the new propagation experiment. These statements distinguish an economic improvement over the restricted class, a sharper global lower bound, and exact global optimality. The original stopped-control objective, adverse comparisons, and complete scalar proof remain intact rather than being removed or relabeled.